

Anup Rai

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SUMMARY

Electronics and Communication Engineering student focused on machine learning, data analytics, and LLM-based systems. Hands-on experience building end-to-end intelligent applications, including retrieval-augmented generation (RAG) pipelines, local LLM-powered question answering systems, and real-time computer vision interfaces. Strong foundation in Python, Scikit-learn, and data processing with practical exposure to embedding-based retrieval and context-aware AI systems. IEEE Best Paper Award recipient (AI ML Track, 2026), with a demonstrated ability to translate research into deployable solutions.

TECHNICAL SKILLS

Programming Languages: Python, C, C++

Machine Learning: Supervised/Unsupervised Learning, K-Means, Scikit-learn, NumPy, Pandas

Probabilistic Models: Bayesian Knowledge Tracing (BKT), Deep Knowledge Tracing (DKT)

Computer Vision: OpenCV, MediaPipe

Tools & Platforms: Git, Jupyter, Google Colab, Unity, Ollama

Other: Graphic Design, Video Editing

PROJECTS

Inclusive Mixed Reality Math Experience Using ML-Based Models

2025 – 2026

Final Year Project

Python, OpenCV, MediaPipe, Scikit-learn

- Built a real-time gesture-controlled tutoring system integrating computer vision and adaptive ML
- Implemented Bayesian Knowledge Tracing (BKT) for adaptive mastery modeling
- Applied clustering techniques to enable personalized learning pathways

MediaPipe Cube Control via Hand Gesture

2026

Computer Vision / HCI

Unity, C#, MediaPipe

- Developed a low-latency real-time hand-tracking interface using MediaPipe integrated with Unity
- Designed modular gesture-state architecture for scalable gesture recognition

Local Letter RAG System

2026

LLM / Retrieval-Augmented Generation (RAG)

Python, LangChain, Ollama

- Built a local Retrieval-Augmented Generation (RAG) pipeline for document-based question answering using embeddings and vector search
- Integrated local LLMs via Ollama to enable offline, privacy-preserving inference without reliance on external APIs
- Designed context-aware retrieval mechanism to improve response relevance and accuracy over naive prompt-based approaches

Smart KYC: Clarity at Every Step

2026

Machine Learning / FinTech UX

Python, Streamlit, NLP

- Built an intelligent KYC assistance system to guide users through document submission with real-time validation and feedback
- Designed a Streamlit-based interface simulating onboarding workflows, improving clarity and reducing user input errors

RESEARCH

DKT-Based AR Framework for Predictive Learning Assessment

2025 – 2026

IEEE ICTP 2026 | Best Paper Award – AI & ML Track

- Designed an adaptive Augmented Reality system using Deep Knowledge Tracing (DKT)
- Developed system architecture following Research-Through-Design (RTD) methodology
- Achieved Accuracy: 0.84 and PR-AUC: 0.82 in real-world classroom deployment

EDUCATION

Nepal Engineering College, Pokhara University
B.E. in Electronics & Communication Engineering
Pokhara Aadarsha Secondary School
Science (Grade 11 – 12)

Bhaktapur, Nepal
2021 – 2026
Pokhara, Nepal
2019 – 2021

CERTIFICATIONS & HONORS

- **Best Paper Award** – IEEE ICTP 2026 (AI & ML Track)
- Data Science with Python – Code IT
- IEEE Student Volunteer